

Software Requirements Specification

for

Hospital Management System

Version 1.0 approved

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1. Introduction

1.1 Purpose

- This document describes the software requirements for the **Hospital Management System (HMS)**.
- It explains what the system will do and what features it must have.
- This document will help developers, stakeholders, and reviewers understand the system clearly.
- The SRS covers the **entire HMS software**, not hardware or medical machines.

1.2 Document Conventions

- Headings are numbered (1, 1.1, 1.2) for easy reading.
- Functional requirements use statements like **“The system shall...”**.
- All requirements are treated as important unless stated otherwise.
- Technical terms are used consistently.

1.3 Intended Audience and Reading Suggestions

This document is useful for:

- **Developers** – to build the system correctly
- **Project managers/reviewers** – to evaluate the system requirements
- **Testers** – to verify system functionality

Reading order:

- First read **Introduction** and **Product Scope** for overview
- Read **Functional Requirements** for system features
- Read **Non-functional Requirements** for performance and security

1.4 Product Scope

- Hospital Management System (HMS) is software to manage hospital activities digitally.
- It replaces manual work with an automated system.

Main goals of HMS:

- Manage patient records easily
- Handle doctor and staff information
- Schedule appointments efficiently
- Maintain billing and payment records
- Store medical history securely

Benefits:

- Reduces paperwork
- Saves time for staff and patients
- Improves data accuracy

- Makes hospital operations smooth and organized

1.5 References

- IEEE 830 – Software Requirements Specification Guidelines
- ISO/IEC/IEEE 29148 – Requirements Engineering Standard
- Software Engineering textbook by Roger S. Pressman
- Internal Project Guidelines Document

2. Overall Description

2.1 Product Perspective

- Hospital Management System (HMS) is a **new, self-contained software product**.
- It is designed to replace traditional manual and paper-based hospital processes.
- The system works as a **centralized platform** for managing hospital operations.
- HMS is not a part of any existing software family.
- It can operate independently but may interact with external systems such as:
 - Payment gateways
 - Laboratory systems
 - Notification services (SMS/Email)

2.2 Product Functions

The major functions of the Hospital Management System include:

- Patient registration and management
- Doctor and staff management
- Appointment scheduling
- Medical record management
- Billing and payment handling
- Pharmacy and medicine management
- User authentication and role-based access control
- Report generation and data viewing

2.3 User Classes and Characteristics

The system will be used by different types of users:

- **Administrator**
 - Manages users, roles, and system settings
 - Has full access to the system
 - Technically skilled
- **Doctor**
 - Views patient records
 - Updates diagnosis and prescriptions
 - Moderate technical knowledge
- **Nurse/Staff**
 - Assists in patient handling

- Updates basic patient information
 - Basic computer skills
- **Receptionist**
 - Registers patients
 - Schedules appointments
 - Limited system access
- **Pharmacist**
 - Manages medicines and prescriptions
 - Updates stock information
- **Patient**
 - Views appointments and medical history
 - Minimal technical knowledge

2.4 Operating Environment

- The system will run as a **web-based application**.
- Supported platforms:
 - Windows, Linux, or macOS
- Supported browsers:
 - Google Chrome, Mozilla Firefox, Microsoft Edge
- Server-side environment:
 - Web server and application server
- Database:
 - Relational database system

2.5 Design and Implementation Constraints

- The system must follow standard software engineering practices.
- Security rules must be implemented to protect patient data.
- Role-based access control must be enforced.
- The system should support concurrent users.
- The database must ensure data consistency and integrity.
- Use of standard web technologies and frameworks.

2.6 User Documentation

The following documents will be provided:

- User manual
- Online help documentation
- Installation and setup guide
- Basic usage tutorials

Documentation will be provided in **digital format**.

2.7 Assumptions and Dependencies

- Users have basic computer and internet knowledge.
- The system will be accessed through a stable internet connection.
- Third-party services (email/SMS, payment gateway) will be available.

- Required hardware and server infrastructure will be provided.
- Changes in hospital rules or policies may require system updates.

3. External Interface Requirements

3.1 User Interfaces

- The system will provide a **web-based graphical user interface (GUI)**.
- Interfaces will be simple, clean, and easy to understand.
- The same layout style will be followed across all screens.

General UI features:

- Login screen for all users
- Dashboard based on user role
- Forms for data entry (patients, appointments, billing, etc.)
- Tables for viewing records
- Common buttons: Save, Update, Delete, Cancel, Search
- Confirmation messages for important actions
- Clear error messages for invalid input

Design standards:

- Consistent fonts and colors
- Proper labels for all input fields
- Mandatory fields clearly marked
- Help or guidance text where required

3.2 Hardware Interfaces

- The system does not directly control any hardware devices.
- It will run on standard computer systems.

Supported hardware:

- Desktop or laptop computers
- Server machine for hosting the application

Optional interfaces:

- Printer for reports and bills
- Barcode scanner (if used by hospital)

All hardware interaction will be handled through the operating system.

3.3 Software Interfaces

The system will interact with the following software components:

- **Operating System**
 - Windows / Linux / macOS
- **Database System**
 - Relational database (e.g., MySQL or PostgreSQL)
 - Stores patient, staff, appointment, and billing data
- **Web Browser**
 - Google Chrome, Mozilla Firefox, Microsoft Edge
- **External Services**
 - Email or SMS service for notifications
 - Payment gateway for online payments (if enabled)

Data exchange will be done using standard formats such as JSON.

3.4 Communications Interfaces

- The system will use standard **web communication protocols**.

Communication methods:

- HTTP/HTTPS for client-server communication
- REST-based APIs for data exchange

Security requirements:

- Secure communication using HTTPS
- Authentication required for all users
- Session management to prevent unauthorized access

Other features:

- Email or SMS notifications for appointments and updates
- Data transfer will occur over a secure network connection

4. System Features

This section describes the major functional features of the Hospital Management System. Each feature explains **what the system does**, **how users interact**, and **what the system must support**.

4.1 User Authentication and Authorization

4.1.1 Description and Priority

- This feature allows users to access the system securely.
- It ensures only authorized users can use specific system functions.
- Access is controlled based on user roles.
- **Priority:** High

4.1.2 Stimulus / Response Sequences

- User opens the login page.
- User enters username and password.
- System verifies login credentials.
- If valid, system redirects user to role-based dashboard.
- If invalid, system displays an error message.

4.1.3 Functional Requirements

- **REQ-4.1.1:** The system shall allow users to log in using valid credentials.
- **REQ-4.1.2:** The system shall restrict access based on assigned user roles.
- **REQ-4.1.3:** The system shall display an error message for invalid login attempts.
- **REQ-4.1.4:** The system shall allow users to log out securely.

4.2 Patient Management

4.2.1 Description and Priority

- This feature manages patient personal and medical details.
- It allows hospital staff to create and maintain patient records.
- **Priority:** High

4.2.2 Stimulus / Response Sequences

- User selects patient management option.
- User enters new patient details or searches existing patient.
- System validates and stores patient data.
- System displays confirmation or patient information.

4.2.3 Functional Requirements

- **REQ-4.2.1:** The system shall allow registration of new patients.
- **REQ-4.2.2:** The system shall allow updating patient information.
- **REQ-4.2.3:** The system shall allow viewing patient medical records.
- **REQ-4.2.4:** The system shall prevent duplicate patient entries.

4.3 Appointment Scheduling

4.3.1 Description and Priority

- This feature manages doctor-patient appointments.
- It helps organize schedules and reduce waiting time.
- **Priority:** High

4.3.2 Stimulus / Response Sequences

- User selects appointment scheduling option.
- User chooses doctor, date, and time.

- System checks doctor availability.
- System confirms or rejects the appointment.

4.3.3 Functional Requirements

- **REQ-4.3.1:** The system shall allow users to book appointments.
- **REQ-4.3.2:** The system shall prevent appointment time conflicts.
- **REQ-4.3.3:** The system shall allow users to view appointment schedules.
- **REQ-4.3.4:** The system shall allow appointment cancellation or rescheduling.

4.4 Billing and Payment Management

4.4.1 Description and Priority

- This feature manages billing and payment details.
- It ensures accurate billing for hospital services.
- **Priority:** Medium

4.4.2 Stimulus / Response Sequences

- User selects billing option.
- System generates bill based on services used.
- User completes payment.
- System updates billing and payment status.

4.4.3 Functional Requirements

- **REQ-4.4.1:** The system shall generate bills for patient services.
- **REQ-4.4.2:** The system shall record payment transactions.
- **REQ-4.4.3:** The system shall display payment history.
- **REQ-4.4.4:** The system shall support online payments if enabled.

4.5 Medical Records Management

4.5.1 Description and Priority

- This feature stores patient diagnosis, prescriptions, and reports.
- It ensures medical data confidentiality and accuracy.
- **Priority:** High

4.5.2 Stimulus / Response Sequences

- Doctor accesses patient record.
- Doctor adds or updates medical details.
- System saves the updated information.

4.5.3 Functional Requirements

- **REQ-4.5.1:** The system shall allow doctors to update medical records.
- **REQ-4.5.2:** The system shall restrict medical record access to authorized users only.

- **REQ-4.5.3:** The system shall maintain a history of medical updates.

5. Other Nonfunctional Requirements

5.1 Performance Requirements

- The system shall respond to user actions within **2–3 seconds** under normal load.
- The system shall support **multiple concurrent users** without performance degradation.
- Patient record retrieval shall complete within **5 seconds**.
- Appointment booking and billing operations shall complete in real time.
- The system shall handle peak usage during hospital working hours smoothly.

5.2 Safety Requirements

- The system shall prevent accidental deletion of critical patient data.
- The system shall require confirmation before performing irreversible actions.
- The system shall maintain data backups to prevent data loss.
- Unauthorized users shall not access or modify sensitive data.
- System failures shall not result in corruption of stored medical records.

5.3 Security Requirements

- The system shall require user authentication before granting access.
- Role-based access control shall be enforced for all users.
- Patient medical records shall be accessible only to authorized personnel.
- Passwords shall be stored in encrypted form.
- All communication between client and server shall use **secure protocols (HTTPS)**.
- The system shall log user activities for security monitoring.

5.4 Software Quality Attributes

- **Usability:**

- The system shall be easy to use with minimal training.
- User interfaces shall be clear and consistent.

- **Reliability:**

- The system shall operate continuously during hospital working hours.
- System crashes and failures shall be minimized.

- **Maintainability:**

- The system shall be modular and easy to update.
- Future enhancements shall be possible without major changes.

- **Portability:**

- The system shall run on different operating systems via a web browser.

- **Scalability:**

- The system shall support future increase in users and data volume.

5.5 Business Rules

- Only administrators can manage users and roles.
- Doctors can update medical records but cannot modify billing details.
- Receptionists can register patients and schedule appointments.
- Patients can view their own records but cannot modify them.
- Billing actions must be linked to valid patient records.

6. Other Requirements

This section includes additional requirements that are not covered in earlier sections.

6.1 Database Requirements

- The system shall use a relational database to store data.
- The database shall store patient, doctor, appointment, billing, and user data.
- Data integrity and consistency shall be maintained at all times.
- Regular database backups shall be supported.

6.2 Legal and Regulatory Requirements

- The system shall ensure confidentiality of patient information.
- Access to medical records shall follow hospital data privacy rules.
- Patient data shall not be shared without proper authorization.

6.3 Data Backup and Recovery

- The system shall support periodic data backups.
- The system shall allow recovery of data in case of system failure.

6.4 Reusability and Extensibility

- The system shall be designed in a modular manner.
- New features can be added with minimal impact on existing modules.

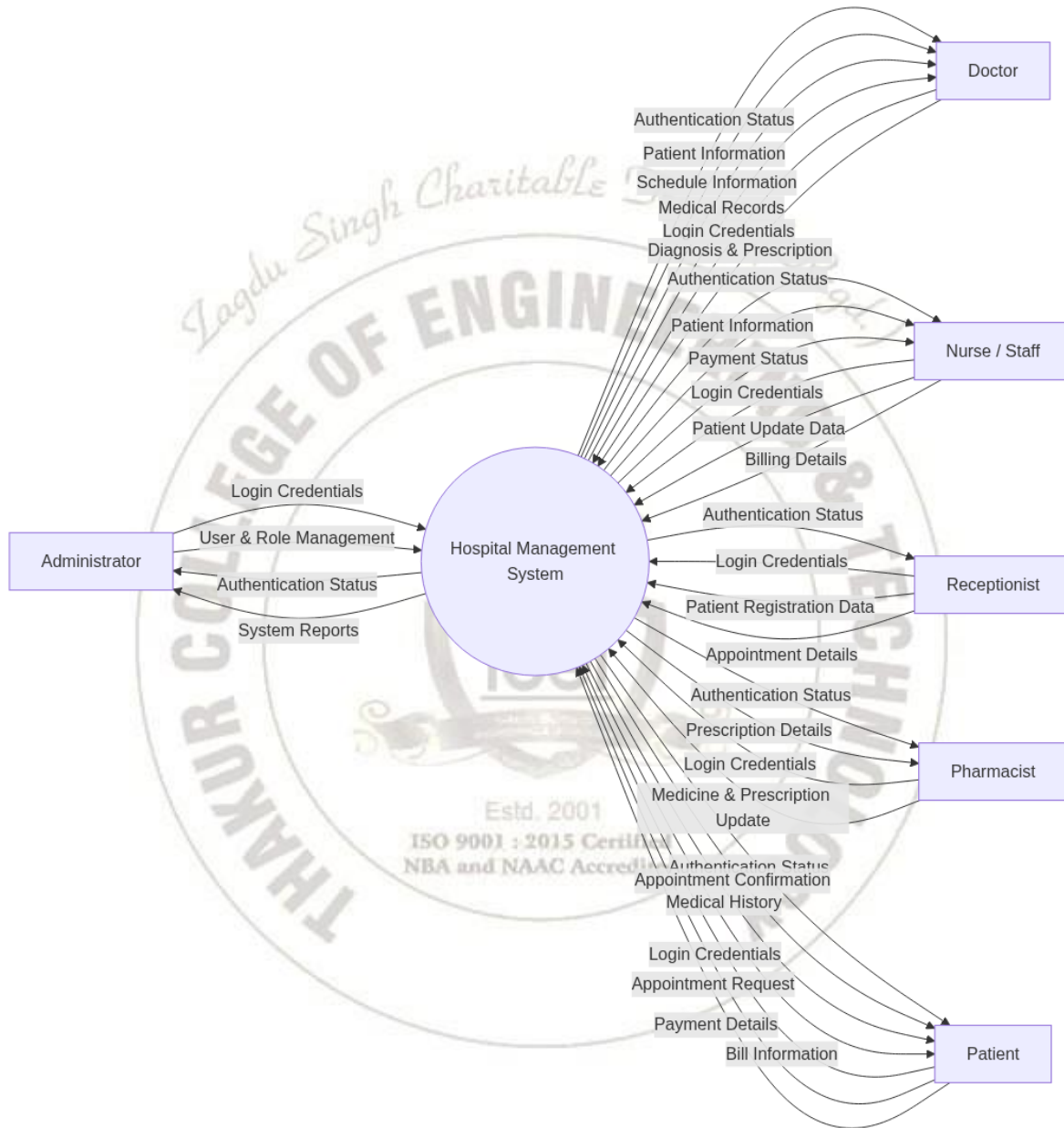
Appendix A: Glossary

Term	Description
HMS	Hospital Management System
GUI	Graphical User Interface
Admin	System administrator
RBAC	Role-Based Access Control
API	Application Programming Interface
HTTPS	Secure Hypertext Transfer Protocol
Database	Structured data storage system

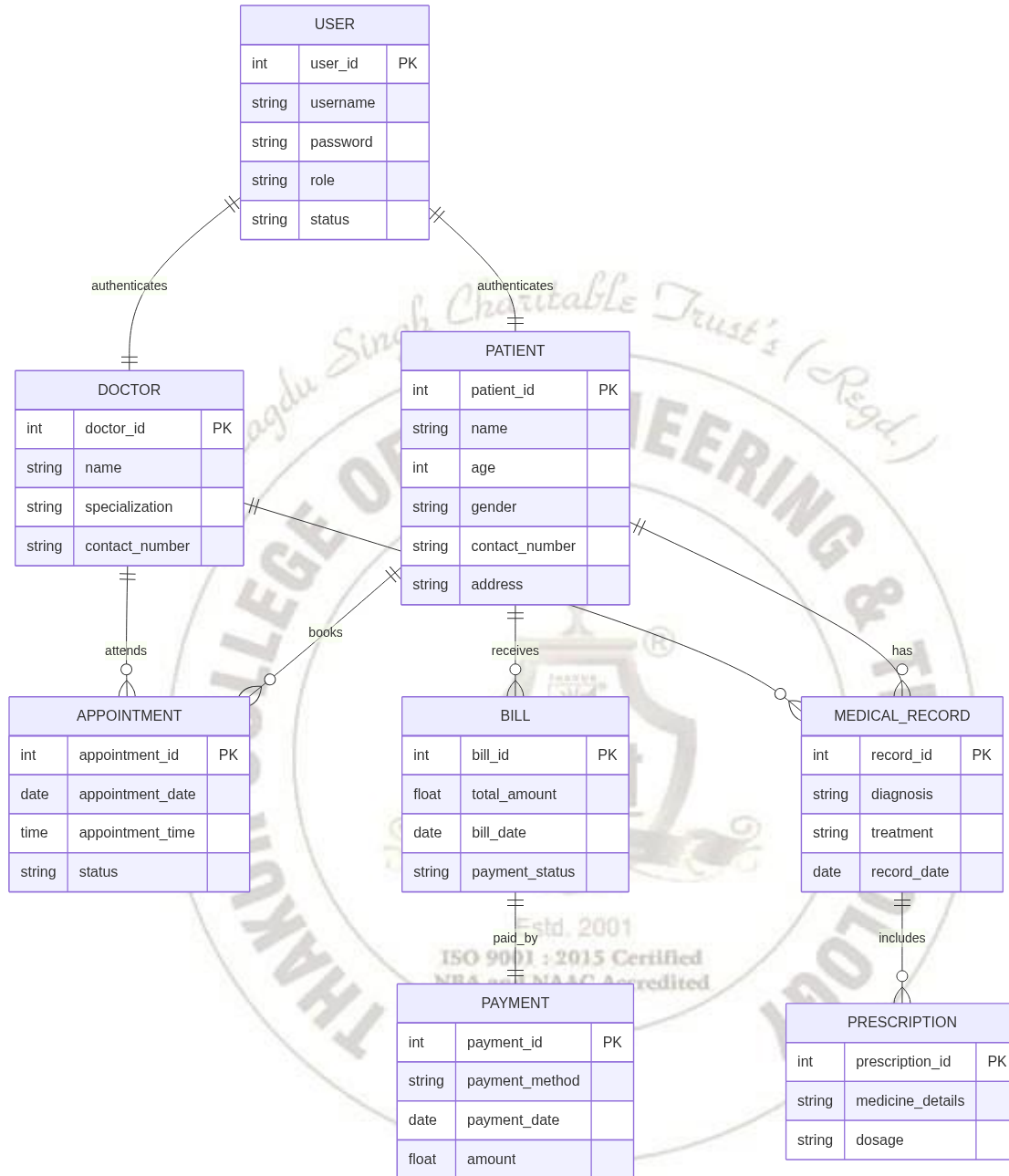
Appendix B: Analysis Models

The following analysis models may be used to describe the system behavior and structure:

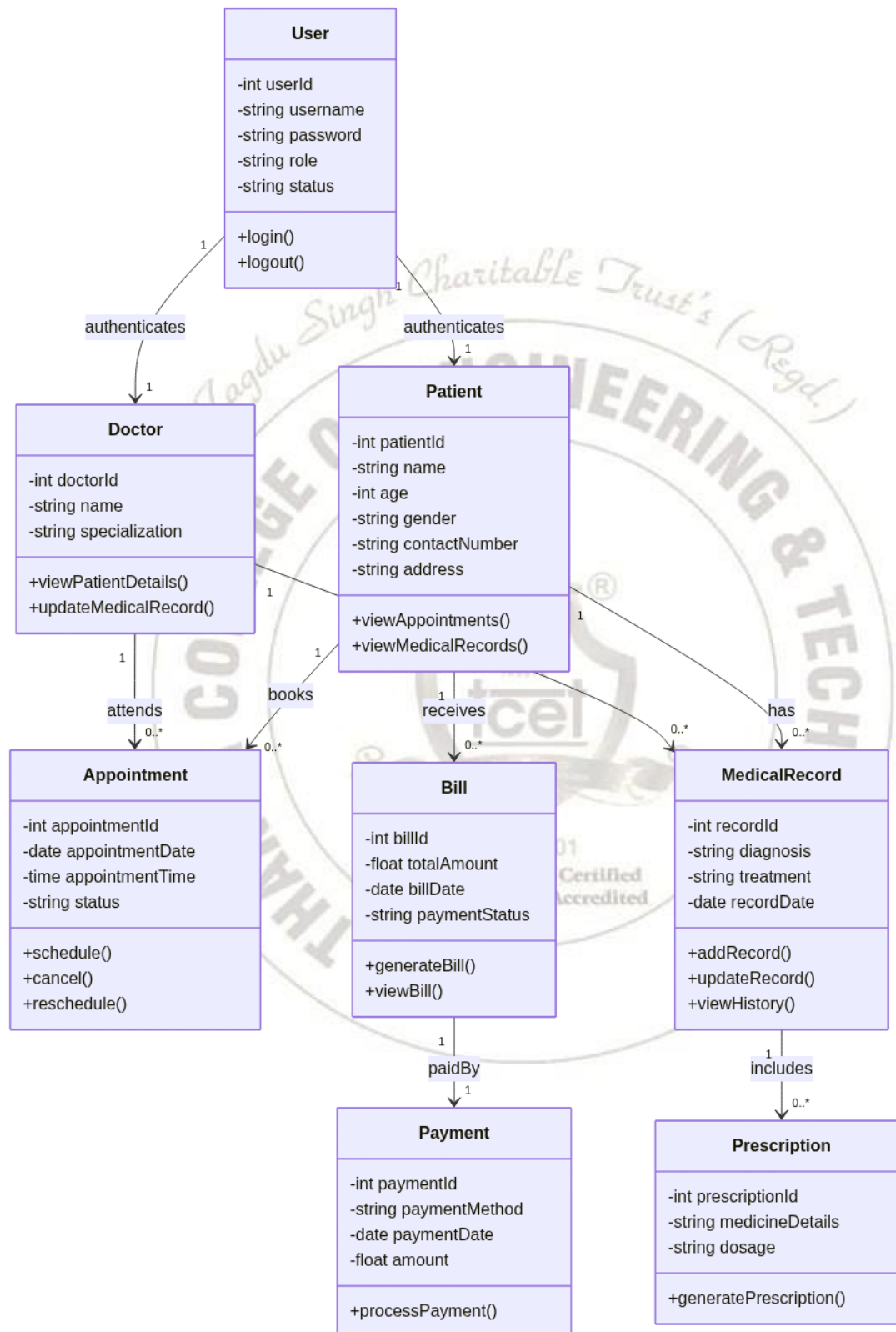
- Data Flow Diagram (DFD)



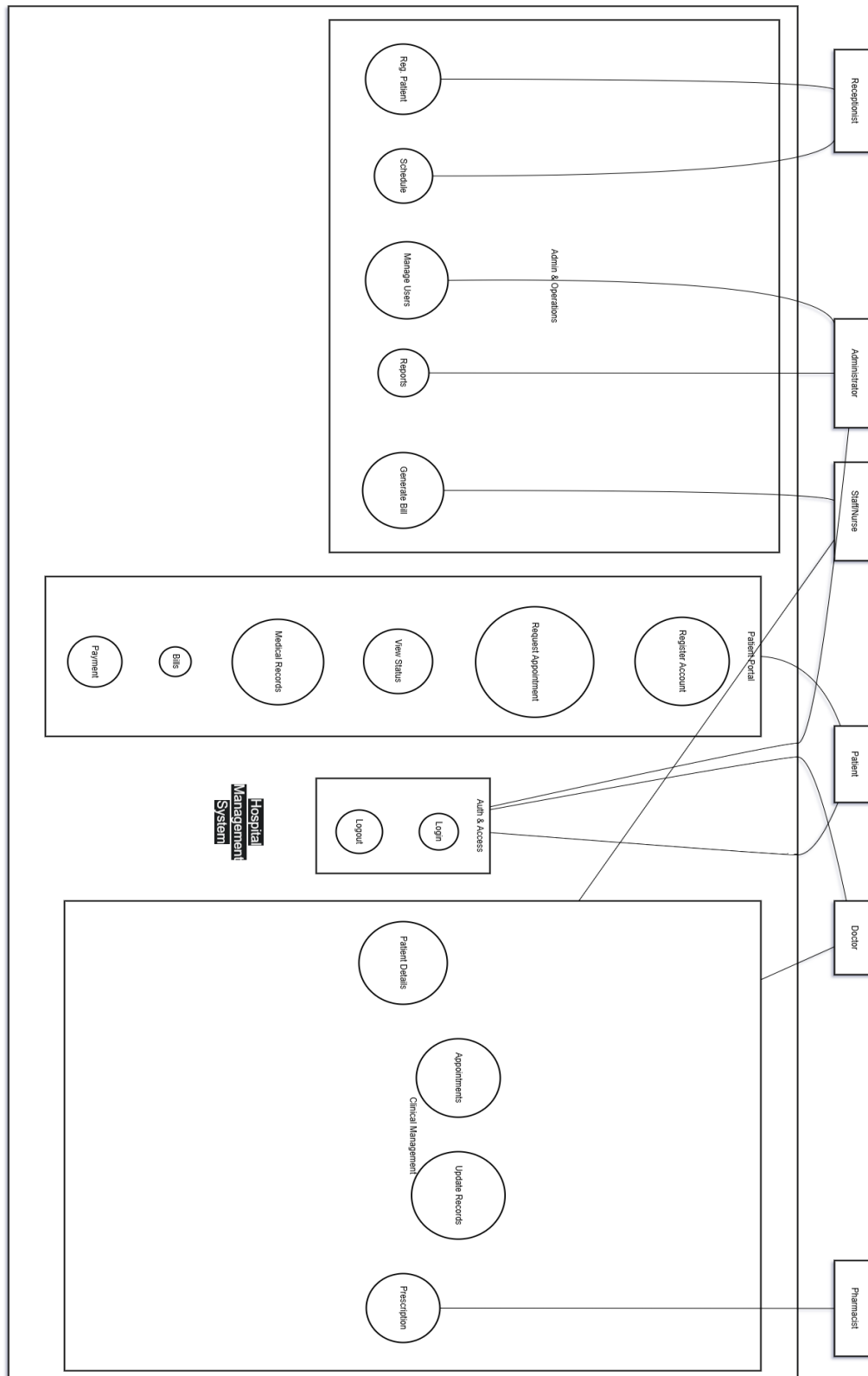
- Entity Relationship Diagram (ERD)



- Class Diagram



- Use Case Diagram



These models help in understanding data flow, system structure, and user interactions.

Appendix C: To Be Determined List

The following items are marked as TBD and may be finalized during later stages of development:

1. Exact database technology to be used
2. External payment gateway details
3. SMS/Email service provider
4. Backup frequency configuration

